Appendix B | Written solutions

Matrix Calculus and Optimization

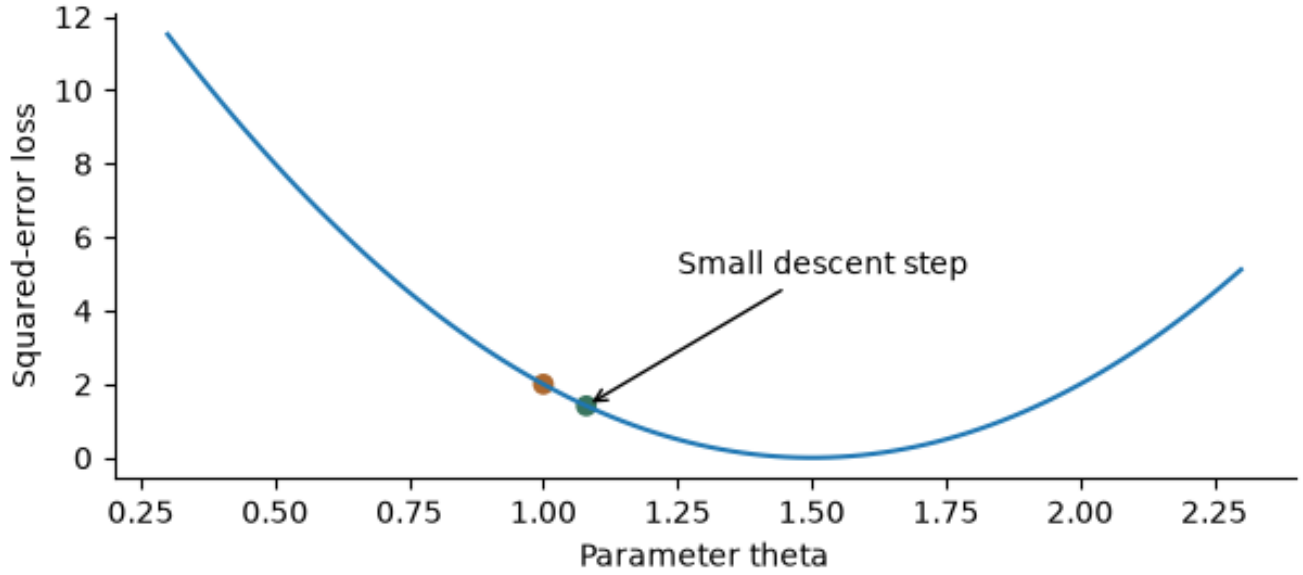
Gradients add along shared paths, token means require token weights, and nonlinear optimizer operations must be applied in the correct order. The examples check derivatives independently and separate adaptive updates from weight decay.

Eight exercises progress from tracing small calculations to diagnosing implementation errors. These are controlled numerical examples, not performance or model-quality benchmarks.

1. Follow a scalar derivative

Problem. For $z = \theta x$, $L = (z - y)^2/2$, take $\theta = 1$, $x = 4$, $y = 6$. Find loss and derivative, and the direction of a small descent step.

Solution. The prediction is 4, the residual is -2, and the loss is 2. The chain rule gives $(z-y)x = (-2)(4) = -8$. Descent subtracts this negative gradient, so it increases the parameter. At learning rate 0.01, the new parameter is 1.08, the prediction is 4.32, and the loss is 1.4112. A very large step could overshoot; the derivative describes a local direction.



2. Differentiate cross-entropy

Problem. For logits $(0, \log 3)$ and the first class as target, compute probabilities, loss, gradient, and the sum of gradient entries.

Solution. The exponentials are 1 and 3, so probabilities are $(1/4, 3/4)$. The correct-class loss is $-\log(1/4) = \log 4$. Subtracting target $(1, 0)$ gives gradient $(-3/4, 3/4)$. Its entries sum to zero: moving both logits by one common constant leaves their differences, probabilities, and loss unchanged. A descent step raises the correct logit and lowers the other.

3. Weight by valid tokens

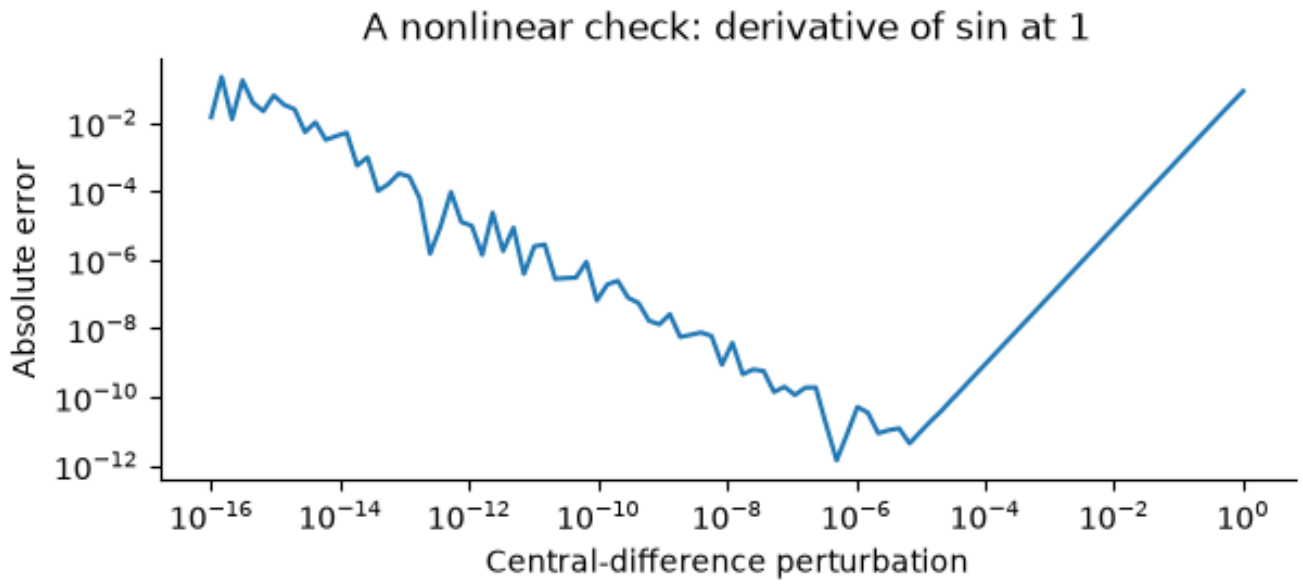
Problem. Two pieces contain two and five valid labels, with mean gradients 4 and -3. Find the combined mean. Does splitting the second piece change it when sums are used?
Solution. The loss-gradient sums are $2(4) = 8$ and $5(-3) = -15$. Divide their total -7 by seven valid labels to obtain -1. Averaging the two means gives 0.5, which incorrectly gives equal weight to unequal token counts. Splitting a piece preserves its total gradient sum and token count, so it preserves the combined result. This argument assumes the same parameters and per-token forward computations.

4. Clip once and separate decay

Problem. Two tensors have gradients $(6, 8)$ and $(0, 0)$. Clip their global norm to 5. Separately, evaluate AdamW for old parameter 2, corrected moments 1 and 4, learning rate 0.1, decay 0.2, negligible epsilon.
Solution. The concatenated gradient has norm 10, so all entries receive multiplier $5/10$. The tensors become $(3, 4)$ and $(0, 0)$. In the separate AdamW calculation, decay contributes $-0.1(0.2)(2) = -0.04$. The adaptive step contributes $-0.1(1)/\sqrt{4} = -0.05$. The new parameter is $2 - 0.04 - 0.05 = 1.91$. Decay is applied to the old weight outside the adaptive division.

5. Check every matrix derivative

Problem. Use $X = ((2, 1), (1, 3))$, $W = ((1, 0, -1), (2, 1, 0))$, and $G = ((1, 0, -1), (0, 2, 1))$. For $L = \sum G_{ij}(XW)_{ij}$, check all weight and input derivatives by float64 central differences at 10^{-6}. Why hold G fixed?
Solution. The fixed array G defines this scalar probe loss. Differentiating G as well would define a different function. The analytic derivatives are $X^\top G = ((2, 2, -1), (1, 6, 2))$ and $GW^\top = ((2, 2), (-1, 2))$. Each central difference perturbs one coordinate, evaluates the same scalar loss twice, and divides by twice the perturbation. This loss is linear in each array when the other is fixed, so central differences have no truncation error in exact arithmetic; floating-point cancellation still remains. A nonlinear example illustrates why the perturbation cannot be arbitrarily tiny.



6. Add shared contributions

Problem. For $Y = XW + b$ with the same G, compute the shared-bias gradient. A second branch contributes ones to the input gradient. Find the total and explain why it is a sum.
Solution. The shared bias affects both rows, so its gradient is their sum: $(1, 2, 0)$. The first branch gives input gradient $((2, 2), (-1, 2))$; adding the second gives $((3, 3), (0, 3))$. Both paths affect the same scalar loss, and the chain rule adds their effects. Averaging would incorrectly halve the derivative unless the loss itself included that averaging factor.

7. Order accumulation and clipping

Problem. Equal-size pieces have scalar mean gradients 10 and -6. Clip at 5. Compare clipping before and after accumulation; repeat with scale 128, and test clipping before unscaling.
Solution. The combined mean is $(10 - 6)/2 = 2$ and remains 2 after clipping. Clipping each piece first gives $(5 - 5)/2 = 0$. With scaled piece means 1280 and -768, sum to 512, unscale to 4, divide by two, then clip to obtain 2. Equivalently, form the scaled mean 256, unscale it, then clip. Clipping that scaled mean at the unscaled threshold 5 and dividing by 128 instead gives 0.0390625, which is wrong. For loss sums with unequal valid counts, normalize by the total count rather than the number of pieces.

8. Trace the first AdamW step

Problem. Start at parameter 2, zero moments, beta values 0.9 and 0.99, learning rate 0.1 and decay 0.2. Compute the first step for gradients 3 and -3, neglecting epsilon.
Solution. For $g = 3$, the moments are $m = 0.1(3) = 0.3$ and $v = 0.01(9) = 0.09$. Dividing by $1 - 0.9$ and $1 - 0.99$ gives corrected moments 3 and 9. Decay contributes -0.04 and the adaptive term contributes -0.1, giving 1.86. For $g = -3$, the first moment and corrected first moment change signs; the second moments remain positive. Decay still contributes -0.04, while the adaptive term is $+0.1$, giving 2.06. On this first scalar step with negligible epsilon, the adaptive magnitude depends on the gradient sign, not its nonzero magnitude; this is not ordinary gradient descent.